C4ISTAR
Weapon System Integration

Signal Officer Advance Course

MAJ AR-JAY S SALAN O-146461 (SC) PA

05 October 2026

Instructor Profile



Name: MAJ AR-JAY S SALAN O-146461 (SC) PA

Address: ISU Compound, Cubag, Cabagan, Isabela

Educational Attainment:

MS Information Technology
BS Information Technology
Computer Hardware Servicing NCII
Signal Officer Advance Course
Signal Officer Basic Course

Previous Assignment:

OG3, ASR, PA
5SBN, ASR, PA
10IB, 1ID, PA
64IB, 1ID, PA

References

a. Philippine Army Manual (PAM) 6-00 Command and Control Communications, and Cyber Systems (2nd Edition) dated 11 December 2017;

b. Philippine Army Manual (PAM) 6-0001 Command and Control, Communications, Computer, Intelligence, Surveillance, Target Acquisition, and Reconnaissance (C4ISTAR);

c. PAM 6-0102 Signal Operations Manual dated 15 April 2022;

d. PAM 6-0102-04 Cyberspace Operations dated 24 February 2020;

TERMINAL LEARNING OBJECTIVES

At the end of the subject, the students will be able to discuss the concept of C4ISTAR Weapon System Integration during signal operations IAW PAM 6 0001. (C3)

ENABLING LEARNING OBJECTIVES

- Determine the key components of C4ISTAR Systems. (C3)
- Discuss the principles of interoperability between C4ISTAR and weapon systems. (C3)
- Present the configuration of C4ISTAR systems. (C3)
- Discuss the integrated C4ISTAR and weapon systems. (C3)

Key components of C4ISTAR

1. Command and Control
2. Communications
3. Computer
4. Intelligence
5. Surveillance
6. Target Acquisition
7. Reconnaissance

What are the key Components of C4ISTAR?

1. Command and Control

The core framework of C4ISTAR. Command is the commander's authority and responsibility to direct forces and employ resources, while control includes the activities and systems used to collect, process, display, store, and disseminate relevant information. The C2 system brings together **networks, people, processes, and command posts** to support decision-making and mission accomplishment

What are the key Components of C4ISTAR?

2. Communications

Provides the reliable links necessary for commanders to issue orders, coordinate forces, and receive feedback. The technologies ranging from **radio, satellite, and microwave to wired communications**, with the appropriate means depending on distance, terrain, and other environmental factors.

What are the key Components of C4ISTAR?

3. Computers

Enable the processing, storage, networking, and rapid transmission of information. Modern military communications have evolved from traditional combat-net radios toward **IP-based radios and computer networks**, allowing transmission not only of voice but also e-mail, GPS coordinates, images, and video.

What are the key Components of C4ISTAR?

4. Intelligence

Converts collected information into useful knowledge for commanders. The manual describes the intelligence cycle as **Planning and Direction → Collection → Processing and Exploitation → Analysis and Production → Dissemination and Integration.**

What are the key Components of C4ISTAR?

5. Surveillance

The systematic observation of areas, persons, activities, objects, or domains—including surface, subsurface, aerospace, and cyberspace—using visual, electronic, photographic, aural, or other means. It contributes information required for situational awareness.

What are the key Components of C4ISTAR?

6. Target Acquisition

Involves **detecting, locating, and identifying a target with sufficient accuracy and detail to permit effective weapons employment**. It links information collection with the ability to produce effects against identified targets.

What are the key Components of C4ISTAR?

7. Reconnaissance

Obtains information about an adversary's activities and resources or about the characteristics of an area through visual observation or other detection techniques.

The PA C4ISTAR manual highlights **UAS** as an important PA ISR asset for increasing situational awareness and supporting acquisition, detection, designation, and battle-damage assessment.

How these components work as one system?

Sensors / ISTAR *Detect, observe, locate & collect*	**Communications + Computers** *Transmit, process, integrate & share*	**Command & Control** *Understand, decide, direct and coordinate*	**Forces / Shooters** *Act and create the desired effect*

The ultimate product is **situational awareness and a Common Operational Picture (COP)** that enables commanders to make timely decisions and effectively employ forces and resources. The C4ISTAR manual specifically describes C4ISTAR operations as synchronizing and integrating **C2 centers, information and communication systems centers/nodes, sensors, assets, and processing, exploitation and dissemination systems** in support of current and future operations.

Principles of interoperability between C4ISTAR and weapon systems

Interoperability between **C4ISTAR and weapon systems** should be understood as the ability of heterogeneous sensors, networks, command-and-control systems, and shooters/weapon platforms to **connect, exchange usable information, and operate together toward a common mission objective**.

Principles of interoperability between C4ISTAR and weapon systems

1. Interoperability through defined Interoperability Points (IOP).
2. Common standards and interoperability profiles.
3. Standardized Service Interoperability Points (SIOP).
4. Interface Control Documents (ICD).
5. Seamless interconnectivity through multiple communication media
6. Open architecture.
7. Standardized networking and data exchange.
8. Security must accompany interoperability.

Principles of interoperability between C4ISTAR and weapon systems

1. **Interoperability through defined Interoperability Points (IOP).**

Different systems do **not necessarily have to use identical internal technologies**. What is important is that the points where different systems connect are standardized.

C4ISTAR Manual explains this using the TACOMS concept: the entire protocol stack is defined at the **Interoperability Points where different networks connect**, while internal network technologies may differ.

Principles of interoperability between C4ISTAR and weapon systems

2. Common standards and interoperability profiles.

C4ISTAR manual uses the **NATO Interoperability Standards and Profiles (NISP)** as a model for establishing technical standards and profiles. Importantly, interoperability encompasses **people, processes, technology, and communication information systems**, rather than equipment alone.

Standards and profiles reduce integration time and risk while allowing existing and future capabilities to work together.

Principles of interoperability between C4ISTAR and weapon systems

3. Standardized Service Interoperability Points (SIOP).

A **Service Interoperability Point** is a focal point within an architecture where distinct systems providing the same service can interoperate.

Interoperability profiles identify such elements as capability requirements, protocols, implementation options, and technical standards.

Principles of interoperability between C4ISTAR and weapon systems

4. Interface Control Documents (ICD).

This is particularly important when connecting C4ISTAR with weapon systems. The PA must use an **Interface Control Document** to achieve interoperability among PA capabilities.

The ICD describes the **system interfaces and communication rules** needed to connect systems, ensure compatibility, and define how the overall system interacts.

Thus, before connecting a sensor, BMS, data link, C2 system, or weapon platform, the organization needs to understand **what information is exchanged, how it is formatted, what interface carries it, and how the receiving system interprets it**.

Principles of interoperability between C4ISTAR and weapon systems

5. Seamless interconnectivity through multiple communication media.

The Army operates an amalgam of IPVPN, satellite communications, combat-net radios, cellular, Wi-Fi, legacy systems, and emerging technologies.

The PA therefore treats seamless communications as a prerequisite for interoperability and interconnectivity. Without it, troops and commanders lose the ability to maintain **Situational Awareness (SA)** and a **Common Operational Picture (COP)**.

Principles of interoperability between C4ISTAR and weapon systems

6. Open architecture.

The PA C4ISTAR Architectural Framework uses an **open architecture**. This facilitates system modification and data exchange and allows different capabilities to be integrated through common communications, internet protocols, and standards.

A major advantage identified is the ability to add new capabilities to existing systems instead of replacing the entire architecture

Principles of interoperability between C4ISTAR and weapon systems

7. Standardized networking and data exchange.

The **TCP/IP** is important because it provides a standardized, interoperable, scalable, reliable, flexible, and extensible framework through which diverse systems and devices can communicate.

At the tactical level, the **data links** provide near-real-time continuous information exchange among ground, air, and surface platforms. These links can carry information about enemy positions, friendly forces, weather, and other operationally relevant data.

Principles of interoperability between C4ISTAR and weapon systems

8. Security must accompany interoperability.

Interoperability does not mean unrestricted connectivity. The manual emphasizes protecting **confidentiality, integrity, and availability** as information moves through C4ISTAR networks.

It identifies measures including authentication, authorization, encryption, network segmentation, secure protocols, access controls, and protection against unauthorized access.

Principles of interoperability between C4ISTAR and weapon systems

Sensors → Decision-Makers → Shooters

Sensors collect information concerning the target, location, and situation. Decision-makers use that information to develop plans and decisions. Shooters then execute the plans, while also providing information back to decision-makers as the operational environment changes.

Interoperability does not require every PA system to be the same. Make different systems capable of working together by standardizing their **points of connection, interfaces, protocols, profiles, and information exchange**. This is the foundation for linking C4ISTAR information to weapon-system employment.

AFP C4ISTAR System Framework

The AFP C4ISTAR System Framework emphasizes the integration of the C4ISTAR platforms developed. It ensures that the reach of the AFP C4ISTAR System components is maximized to provide COP/SA to the military decision-makers.

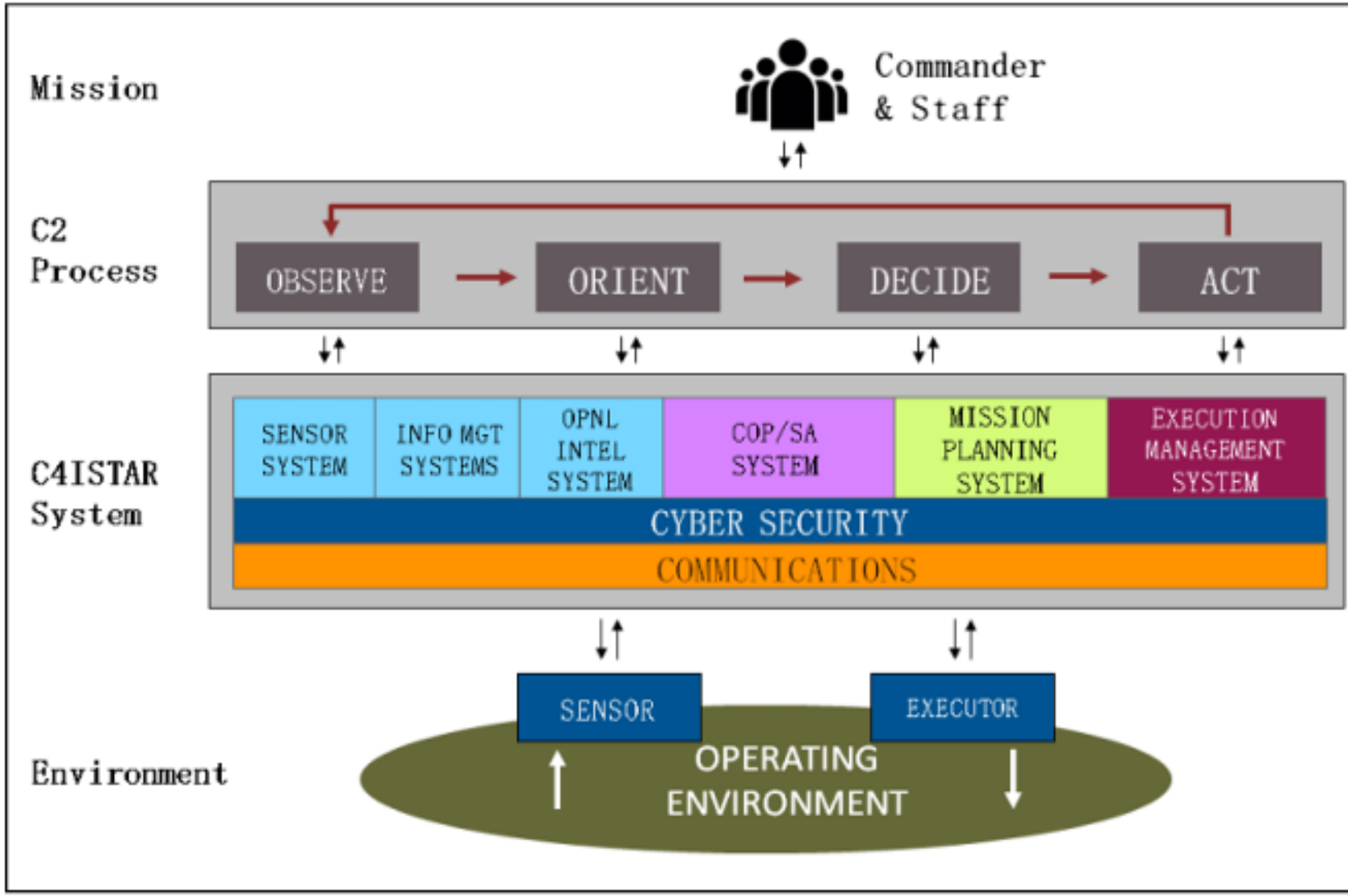


Figure 2.5 – AFP C4ISTAR System Framework
(Source: AFPM 6-07, 2020)

Decide, Detect, Deliver, and Assess (D3A) Process

Provides a systematic approach to the targeting process, helping military commanders make informed decisions and optimize the use of resources.

An integration process during execution, the application enables the staff to use the targeting products to facilitate operations.

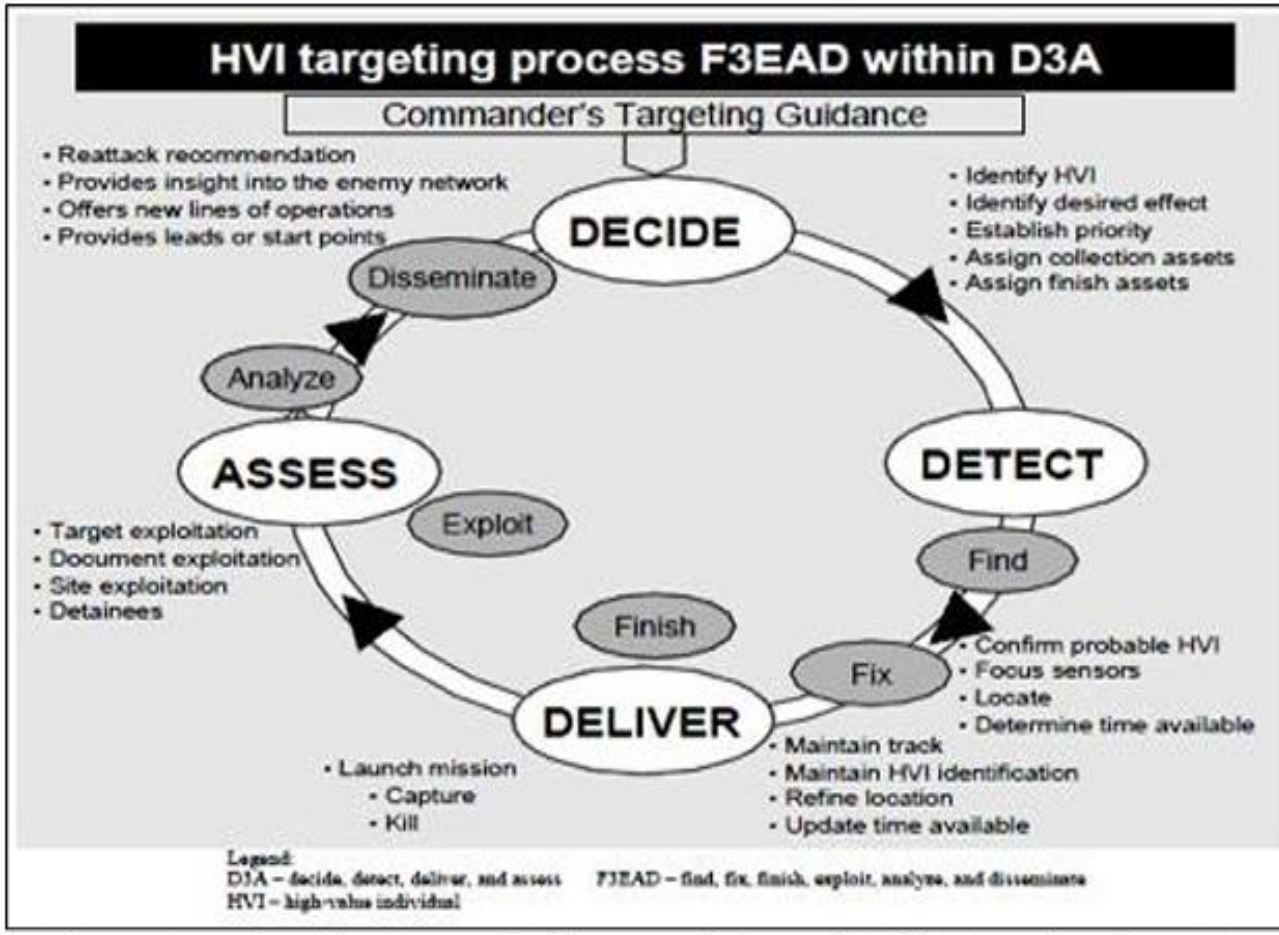


Figure 2.6 – Decide, Detect, Deliver, and Assess (D3A) Targeting Functions
(Source: FM 3-60, 2023)

PA C4ISTAR Architectural Framework

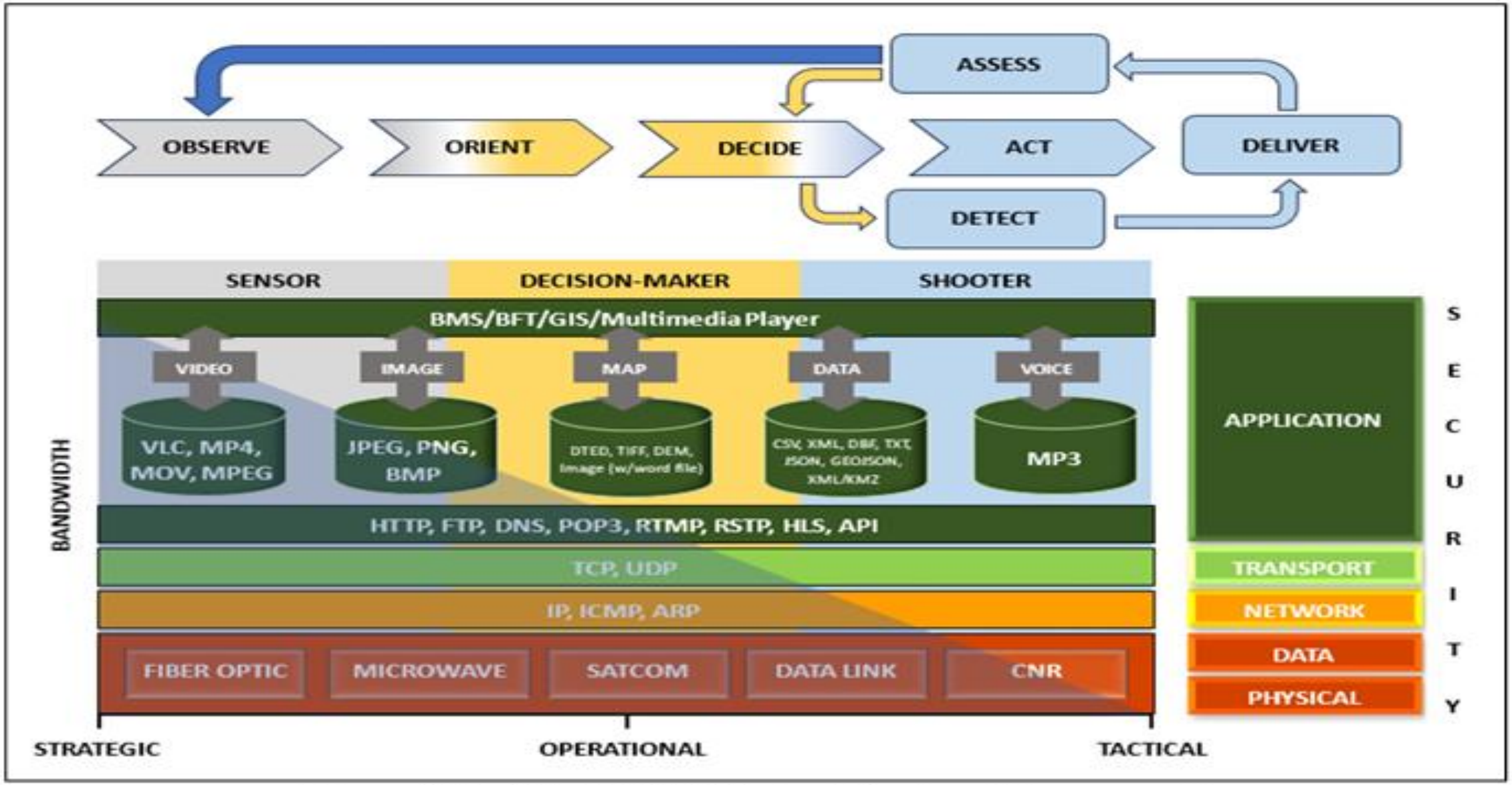


Figure 2.7 – PA C4ISTAR Architectural Framework

PA C4ISTAR Framework

The PA-C4ISTAR Architectural Framework represents the relationship of four underlying components: the PA Core processes in the context of its warfighting function, the Operational Environment, the actors, and the Information Structure.

The PA-C4ISTAR supports the PA's warfighting function in combined and joint operations as well as in a coalition environment. The C2 process depicted in its architectural framework represents a process used by the AFP and other MS C4ISTAR systems– the observe, orient, decide, and act (OODA) loop, reinforced with the D3A methodology.

PA-C4ISTAR MILESTONES



A project initiated by OG3, PA together with OG2, OG6 and AMIC

2005



2007

PA C4ISTAR

Conceptualization of the PA C4ISTAR

2012



NOC

Enhancement of Network Operation Center

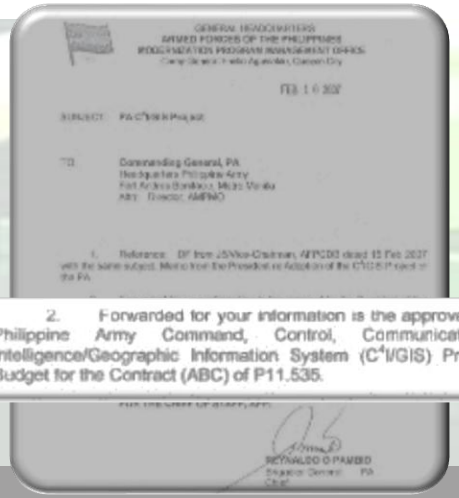
2014

AV RADIOS

Delivery of AV Radios



2015



2. Forwarded for your information is the approval by the President of the Philippine Army Command, Control, Communications, Computer and Intelligence/Geographic Information System (C4I/GIS) Project with an Approved Budget for the Contract (ABC) of P11.535.

PA C4I/GIS I

GHQ approved the adoption of the PAC4I/GIS



PA C4I/GIS II

GHQ approved the PAC4I/GIS II



MCC1

Operationalization of PA's first Mobile Command Center

2014 2015 2018

PA-C4ISTAR MILESTONES



AOC

Enhancement of the AOC Facility to replicate the AFPCC



TACTICAL RADIOS



Delivery of various Combat Net Radios

AV RADIOS

NFS

Procurement of a Night Fighting System



2016

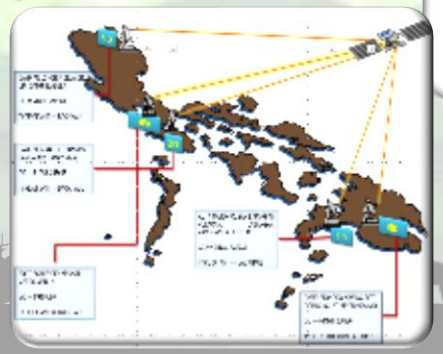


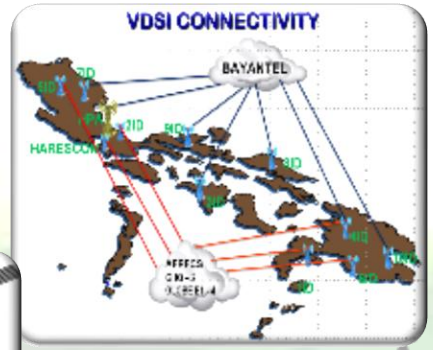
MCC2

Operationalization of PA's second Mobile Command Center

VDSI

Implementation of the Voice and Data System Integration Project





VSAT

Enhanced the voice and data connectivity through VSAT

2017



SATPHONE

PAWAN



BGAN

Enhancement of PA Communications System



PA-C4ISTAR MILESTONES



TOCs

Enhancement of Infantry Division and Bde TOCs

Operationalization of additional two (2) MCCs

LRR MCC





525 MCC





BFT TRACKERS

Procurement of GPS Trackers for BFT purposes

2017

PA VIRTUALIZATION

Enhancement of the PA Data Center Through MITHI





PA EVENTS REPORTING SYSTEM

Development of the Army Events Reporting System integrated with PA GIS

2017



4ID MCC

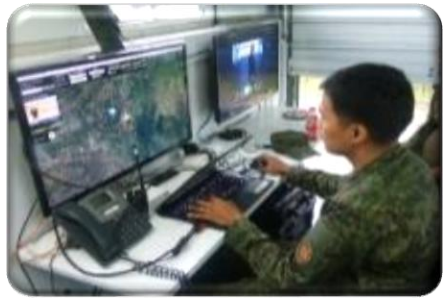
Operationalization of additional two (2) MCCs



MID MCC

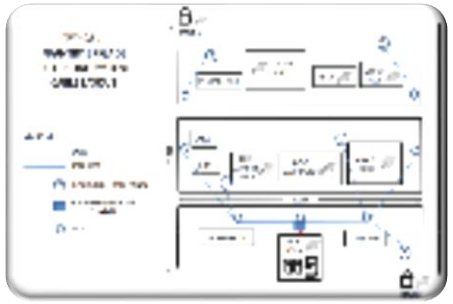
2014 2015 2018

PA-C4ISTAR MILESTONES



NETB BFTS

Development of a homegrown Standalone BFT System for Harris



BDE TEL SYS

Implementation of Bde Telephone System

MCC 3

Operationalization of an additional MCC



2018



PA CYBER LAB

Establishment of PA Cyber defense capabilities thru MITHI



THERMAL IMAGING DEVICE

Procurement of Thermal Imaging Devices

2019



BN TEL SYS

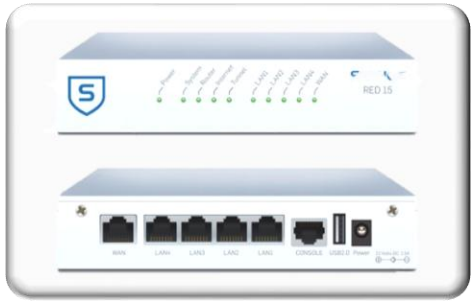
Implementation of Bn Telephone System



IS/EMCS & SOC

Enhancement of PA Mini Security Operation Center and its Information Sharing Portal

PA-C4ISTAR MILESTONES



DBTOCS

Pushed PANET connectivity to the brigade level using VPN Technology



MP Radio

Delivery of HH and MP Radios



HH Radio



MCC FOR ASR, PA/ 11D, PA/8ID, PA

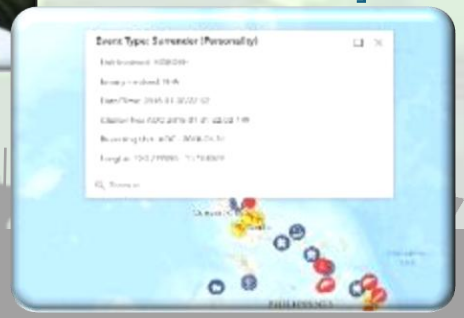
Operationalization of additional three (3) MCCs

2019



PA CYBER RANGE

Developed the AFP's 1st Cyber Range System



INTEGRATION OPNS & INTEL GIS

Integrated the Operations and Intel data into one map.

2020

Additional 30,000 Licensed Mailbox



P25 UHF Radio

Delivery of P25 UHF Radios



BFTS for 3ID, PA/ 4ID, PA/ 11ID, PA

Implementation of BFT System for additional three (3) units

2014 2015 2018

PA-C4ISTAR MILESTONES



PA UAS

Delivery of UAV as part 7 Lots of the PA C4ISTAR Project



DIRECTION FINDER



THERMAL IMAGER



JAMMER



SFLD



SUGV

2021



VSAT

Deployed VSAT to Forward Operating Bases

2022



MANET RADIO



P25 RADIO







PA-C4ISTAR MILESTONES



MCC for 6 PAMUs

PLDT IPVPN





PA C4ISTAR at 6ID



2022

BFTS Version 2



BFTS Version 2 Upgrade at 3ID and 6ID

BFTS Version 2 Implementation at 5ID and 8ID

2023



PA C4ISTAR Integration at 6ID

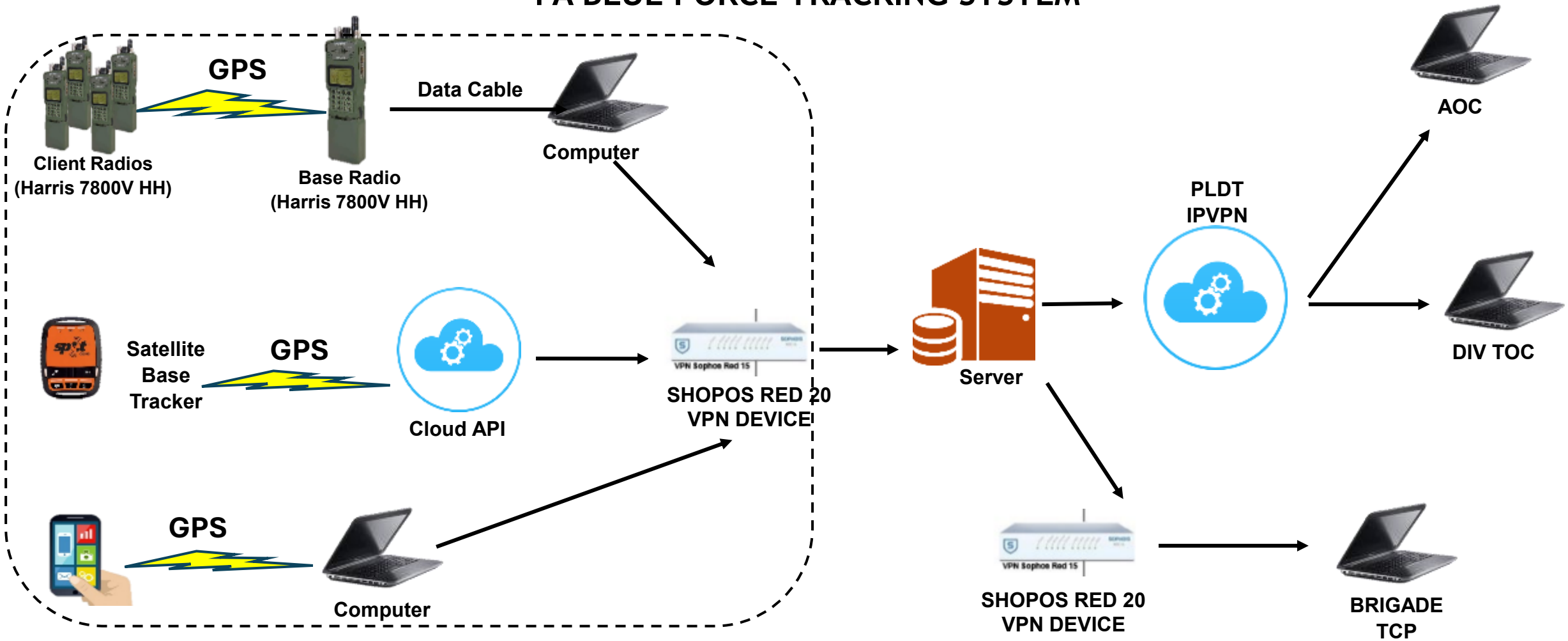




2014 2015 2018

PA C4ISTAR SYSTEM

PA BLUE FORCE TRACKING SYSTEM



2014 2015 2018

PA-C4ISTAR SYSTEM

PA-C4ISTAR SYSTEM AT 6ID (ELBIT SYSTEMS)

SENSORS

MANET RADIO













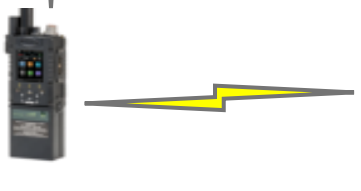






BRIGADE TOC

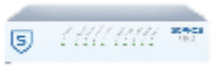




MANET BASE
MANET HH





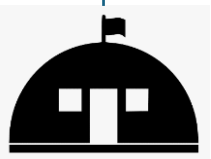


PANET

SHOPOS RED



AOC



DIVISION TOC

PA C4ISTAR SYSTEM

DIVISION TOC



ARMY OPERATIONS CENTER



End of Presentation